

Size statistics of cumulus cloud populations in large-eddy simulations

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How many clouds of each size are there, and which sizes carry the cloud cover and the vertical mass flux?



Aims

- 1 Test LES cloud populations against observed ones, with the algorithm of satellite studies.
- 2 Find a universal form of the cloud size density.
- 3 Find which sizes dominate cover and mass flux.

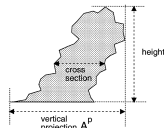
Simulations: KNMI LES

$\Delta x = \Delta y = 50$ m, $\Delta z = 40$ m, 6.4 km domain;
3D fields sampled every 5 min.

BOMEX	ocean, trade-wind cumulus, steady	36,776 clouds
SCMS	Florida, land, deepening layer	20,268 clouds
ARM SGP	Oklahoma, land, diurnal cycle	35,137 clouds

Cloud statistics

Cloud n : connected saturated grid boxes, sized by the projected area A_n^p (right).



$$\ell_n = \sqrt{A_n^p}, \quad \mathcal{N}(l) = a l^b$$

$$\alpha^p(l) = \frac{l^2 \mathcal{N}(l)}{L_x L_y}, \quad \mu(l) = \overline{\alpha(l, z) w(l, z)}$$

$\mathcal{N}$: clouds per unit size. α^p : their share of the projected cover. μ : their share of the mass flux.

LES clouds are measured exactly as satellites measure real ones

A power law with a scale break, collapsed by one length

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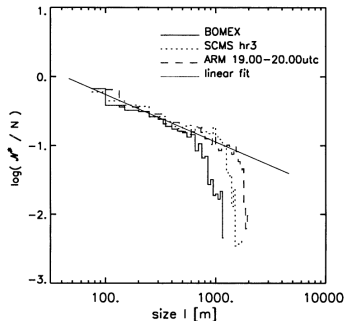


Fig. 4a: $\mathcal{N}^*/N$ for the three cases; fit $1.121 - 0.70 \log l$ below the break.

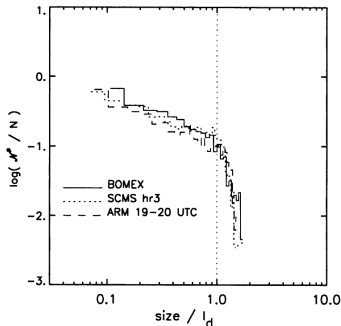


Fig. 6: the same densities against l/l_d ; dotted line at the break.

Below the break

$b = -1.70$ over about one decade, from the grid spacing to ℓ_d .

Observed: mean $b = -1.98$ (Benner & Curry 1998).

Scale break ℓ_d

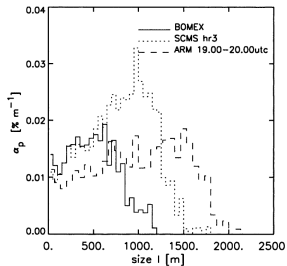
BOMEX	700 m
SCMS	1,050 m
ARM, 15-16 UTC	400 m
ARM, 19-20 UTC	1,250 m

More sampled clouds: the same break.

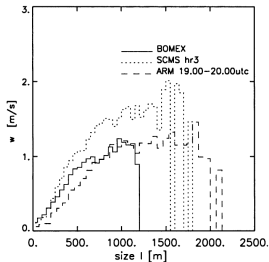
One exponent fits every case; only the break ℓ_d changes

Intermediate clouds carry the cover and the mass flux

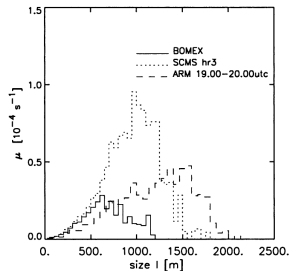
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(a) projected cover $\alpha^p(l)$



(b) cloud-mean $w(l)$



(c) mass flux $\mu(l)$

Robustness

Grid 25–100 m:
same dominant
and largest sizes.

Domain 3.2 km:
the largest clouds
are missing.

Shear $\times 0, 1, 2$:
larger ℓ_d , but the
same total mass
flux.

Pure power law: $\alpha^p \propto l^{b+2}$, so the smallest clouds would dominate if $b < -2$, the largest if $b > -2$. Here $b = -1.70$ below ℓ_d and $b < -2$ above: α^p peaks at the break.

$\mu = \alpha w$ peaks at larger sizes still, since w grows with l . The smallest clouds carry almost no mass flux.

The dominant size sits at the break, far above the grid scale, for cover and for transport

Conclusions

LES reproduces natural populations: a power law below the break, a distinct break above.

Support for the observed power laws and breaks; no evidence for exponential or lognormal forms.

A universal form, with the break ℓ_d as the only case-dependent parameter.

Break plus exponent give an intermediate dominant size, in the cover and in the mass flux.

The dominant clouds sit well above the grid scale; the smallest carry almost no transport.

Limits

One decade of scaling only; 1–2 m grids would be needed for another.

What sets the break is unexplained; the (sub)cloud-layer depth and the shear move it.

Too small a domain truncates the largest clouds.

Three shallow-cumulus cases, compared with published exponents, not the same scenes.

The disagreement with Brown (1999b) is not settled.

Snapshots only, so no cloud life cycle; no uncertainty quoted on b .

Realistic size statistics, one free length, and an unexplained scale break